# EAST WEST UNIVERSITY

Department of Mathematical and Physical Sciences

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**LABORATORY REPORT**

## CHE109 - ENGINEERING CHEMISTRY

### Experiment No: 04

EXPERIMENT TITLE

**Estimation of Ferrous Ion ($Fe^{2+}$) by a Standard $KMnO_4$ Solution**

**SUBMITTED TO**


**Dr. Zubair Hasan**

Associate Professor

Department of Mathematical and Physical Science


**SUBMITTED BY**


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|---|---|
| **Name** | Akayed Hossain |
| **ID** | 2026-2-60-325 |
| **Section** | 01 |
| **Group** | **03** |
| **Department** | **CSE** |


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|---|---|
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**Experiment 4: Estimation of Ferrous ion ($Fe^{2+}$) by a Standard $KMnO_4$ Solution**

## Objective

The aim of this experiment was to determine the concentration of ferrous ions ($Fe^{2+}$) in a supplied solution by titration with standardized potassium permanganate ($KMnO_4$) solution.

## Introduction and Principle

Determination of the concentration of ferrous and ferric ions is essential for understanding the composition of a sample. This is particularly important in analytical chemistry and industries where precise knowledge of elemental concentrations is crucial. In acidic medium, permanganate oxidizes $Fe^{2+}$ to $Fe^{3+}$ while it is reduced to $Mn^{2+}$. The first permanent light pink colour indicates the endpoint.

## Overall reaction

$2KMnO_4 + 10FeSO_4 + 8H_2SO_4 \rightarrow 2MnSO_4 + 5Fe_2(SO_4)_3 + K_2SO_4 + 8H_2O$

(a) $10Fe^{2+} \rightarrow 10Fe^{3+} + 10e^-$ (Oxidation half reaction)

(b) $2MnO_4^- + 16H^+ + 10e^- \rightarrow 2Mn^{2+} + 8H_2O$ (Reduction half reaction)

## Chemicals and Apparatus

**Chemicals:** $H_2C_2O_4$, $KMnO_4$, $FeSO_4.7H_2O$, $H_2SO_4$, etc.

**Apparatus:** Conical Flask, Volumetric Flask, Burette, Dropper, Funnel, Pipette Filler, etc.

## Procedure

a) Standardization of $KMnO_4$ solution by standard $H_2C_2O_4$ solution as expt. 3.

b) Determination of $Fe^{2+}$ in a supplied solution by the standard $KMnO_4$ solution.

1. Pipette out 10 mL of the Ferrous Sulfate Heptahydrate solution in a conical flask.
2. Add 2 mL 1M/2N $H_2SO_4$ into conical flask.
3. Titrate the solution with the standard $KMnO_4$ solution.
4. The end point is indicated by the first appearance of permanent light pink color as expt. 3.
5. Repeat the titration at least 3 times.

## Experimental Data

Titration of $Fe^{2+}$ solution with $KMnO_4$ solution:

| Observation | Initial burette reading (mL) | Final burette reading (mL) | Difference (mL) |
|---|---|---|---|
| 1 | 1.0 | 10.9 | 9.9 |
| 2 | 10.9 | 20.8 | 9.9 |
| 3 | 20.8 | 30.9 | 10.1 |
| Average | — | — | 9.967 |

## Calculations

According to reaction, 2 moles $KMnO_4 \equiv 10$ moles $FeSO_4$, i.e. 1 mol $KMnO_4 \equiv 5$ moles $FeSO_4$

Now, $x \times V(KMnO_4) \times M(KMnO_4) = y \times V(FeSO_4) \times M(FeSO_4)$

## Calculation

$$M(FeSO_4) = \frac{x \times V(KMnO_4) \times M(KMnO_4)}{y \times V(FeSO_4)}$$

$= (5 \times 9.967 \times 0.02) / (1 \times 10) = 0.09967 \text{ mol } L^{-1}$

## Result

Amount of $Fe^{2+}$ ions in 1000 mL of supplied iron solution is 5.565 g.

## Theoretical Value Calculation and Percentage Error

Therefore, 1000 mL $FeSO_4$ solution contain 0.09967 mol $FeSO_4/Fe^{2+}$. The experimental amount is $0.09967 \times 55.84 = 5.565$ g $Fe^{2+}$. The theoretical amount is $0.1 \times 55.84 = 5.584$ g $Fe^{2+}$.

$$\text{Percentage (\%) of Error} = \left|\frac{(\text{Known value} - \text{Experimental value})}{\text{Known value}}\right| \times 100$$

$$= \left|\frac{(5.584 - 5.565)}{5.584}\right| \times 100 \quad = 0.34\%$$

## Answers to Practice Questions

### 1. What do you mean by redox titration?

Answer: A redox titration is a volumetric analysis in which the analyte and titrant react through oxidation and reduction, involving transfer of electrons.

### 2. How will you prepare a 0.003 M 350 mL $KMnO_4$ solution?

Answer: W = M × molar mass × V/1000 = 0.003 × 158.04 × 350/1000 = 0.166 g. Weigh 0.166 g $KMnO_4$, dissolve it in distilled water, transfer to a 350 mL volumetric flask, make up to the mark, and mix.

### 3. What do you mean by oxidant and reductant?

Answer: An oxidant accepts electrons and is reduced. A reductant donates electrons and is oxidized.

### 4. Calculate the equivalent weight of $KMnO_4$.

Answer: In acidic medium, Mn changes from +7 to +2, so n = 5. Equivalent weight = molar mass/n = 158.04/5 = 31.61 g $equiv^{-1}$.

### 5. Discuss the precautions of the above experiment.

Answer: Use clean glassware and rinse the burette with $KMnO_4$ solution and the pipette with $FeSO_4$ solution. Add dilute $H_2SO_4$ to the flask, read the burette at eye level, swirl continuously, and add $KMnO_4$ dropwise near the endpoint. Stop at the first permanent light pink colour and repeat to obtain concordant readings.